

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration,Data transformation and data Reduction , for effective data preparation (BL3,BL4)

QUESTIONS: 5

Total Marks:50

Annexure B
Analytical Problem Solving

Criteria	Problem Understanding (2)	Method Selection (2)	Solution Process (2.5)	Accuracy of Calculation (2)	Interpretation of Results (1.5)	Total(10)
Weightage	20%	20%	25%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I S.Krishna Sagar Reddy(192525384) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:S.Krishna Sagar Reddy

RUBRICS FOR CALCULATION:

Numerical Problems

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Annexure B
Analytical Problem Solving

1. Data Cleaning using Duplicate Detection (Patient Registration) (5 Marks)

The following patient IDs were extracted from a registration database.

P101, P102, P105, P103, P105, P107, P108, P103, P110, P112

Perform the following:

1. Identify duplicate records.
2. Remove duplicate entries.
3. Determine the number of unique patients.

2. Covariance Analysis between Treatment Cost and Hospital Stay (10 Marks)

The following data represent treatment cost (₹ thousand) and hospital stay (days).

Treatment Cost (x) Hospital Stay (y)

20	3
25	4
30	5
35	6
40	8

Perform the following:

1. Compute the mean of x and y.
2. Calculate the covariance.
3. Interpret the relationship between treatment cost and duration of stay.

3. Outlier Detection using Z-Score (Patient Recovery Time) (10 Marks)

Recovery time (days) for patients after surgery is recorded as:

8, 9, 10, 10, 11, 12, 12, 13, 14, 15, 35

Perform the following:

1. Compute the mean.
2. Compute the standard deviation.
3. Calculate the z-score for each observation.
4. Identify the outlier(s) using the criterion $|z| > 3$.

4. Data Discretization using Equal-Width Binning (Blood Glucose Levels) (5 Marks)

A clinic records fasting blood glucose levels (mg/dL).

Sorted Data:

75, 82, 88, 92, 95, 99, 104, 110, 118, 122, 130, 145

Partition the data into 3 equal-width bins.

Replace each observation by the nearest bin boundary.

Write the transformed dataset.

5. Standardization of Insurance Claim Amounts (10 Marks)

An insurance company records claim amounts (₹ thousand):

12, 15, 18, 20, 22, 25, 28, 30, 35, 40

Perform the following:

- 1. Compute the mean.**
- 2. Compute the standard deviation.**
- 3. Calculate the z-score for every observation.**
- 4. Identify claims that are above one standard deviation from the mean.**

Q1: Data Cleaning using Duplicate Detection (Patient Registration)

Step 1: Original Records with Duplicate Flag

S.No	Patient ID	Occurrence Count (in list)	Duplicate? (Y/N)
1	P101	1	No
2	P102	1	No
3	P105	2	Yes
4	P103	2	Yes
5	P105	2	Yes
6	P107	1	No
7	P108	1	No
8	P103	2	Yes
9	P110	1	No
10	P112	1	No

Step 2: Unique Patient List (duplicates removed)

S.No	Unique Patient ID
1	P101
2	P102
3	P105
4	P103
5	P107
6	P108
7	P110
8	P112

Step 3: Summary

Total Records Extracted:	10
Duplicate ID Values Found:	P105, P103
Number of Duplicate (extra) entries	2
Number of Unique Patients:	8

Q2: Covariance Analysis between Treatment Cost and Hospital Stay

S.No	reatment Cost x (₹ '000)	ospital Stay y (day)	x - mean(x)	y - mean(y)	(x-mean x) * (y-mean y)
1	20	3	-10	-2.2	22
2	25	4	-5	-1.2	6
3	30	5	0	-0.2	0
4	35	6	5	0.8	4
5	40	8	10	2.8	28

Sum ->

60

Mean (x, y) ->

30

5.2

Covariance Calculation

Population Covariance $\text{Cov}(x,y) = \Sigma[(x-\bar{x})(y-\bar{y})] / n$

12

Sample Covariance $\text{Cov}(x,y) = \Sigma[(x-\bar{x})(y-\bar{y})] / (n-1)$

15

Interpretation

The covariance between treatment cost and hospital stay is positive (Cov $\approx$ +15, sample basis / +12, population basis). This indicates a positive relationship: as the treatment cost increases, the duration of hospital stay also tends to increase. Patients with higher treatment costs are generally associated with longer hospital stays, which is clinically intuitive since more severe/complex cases require both longer stays and more expensive treatment.

Q3: Outlier Detection using Z-Score (Patient Recovery Time)

S.No	Recovery Time (days)	x - mean	(x-mean)^2	Z-score = (x-mean)/std
1	8	-5.545454545	30.75206612	-0.7837277069
2	9	-4.545454545	20.66115702	-0.6423997597
3	10	-3.545454545	12.57024793	-0.5010718126
4	10	-3.545454545	12.57024793	-0.5010718126
5	11	-2.545454545	6.479338843	-0.3597438655
6	12	-1.545454545	2.388429752	-0.2184159183
7	12	-1.545454545	2.388429752	-0.2184159183
8	13	-0.5454545455	0.2975206612	-0.07708797117
9	14	0.4545454545	0.2066115702	0.06423997597
10	15	1.454545455	2.115702479	0.2055679231
11	35	21.45454545	460.2975207	3.032126866

Sum of squares ->

550.7272727

Mean: 13.54545455

Standard Deviation (popul 7.075741354

Outlier Identification ($|z| > 3$)

Observation	Z-score	Outlier ($ z > 3$)
35	3.032126866	Yes - Outlier

Conclusion: Only the value 35 has $|z| > 3$, so it is identified as the sole outlier in the dataset.

Q4: Data Discretization using Equal-Width Binning (Blood Glucose

Bin Width Calculation

Minimum v **75**
Maximum **145**
Range (M **70**
Number of **3**
Bin width (**23.33333333**

Bin Boundaries

Bin	Lower Boundary	Upper Boundary
Bin 1	75	98.33333333
Bin 2	98.33333333	121.6666667
Bin 3	121.6666667	145

Transformed Dataset (Smoothing by Bin Boundaries — each value replaced by nearest

S.No	Original Value	Assigned Bin	Nearest Lower Edge Dis	Nearest Upper Edge Dis	Transformed Value
1	75	Bin 1	0	23.33333333	75
2	82	Bin 1	7	16.33333333	75
3	88	Bin 1	13	10.33333333	98.33333333
4	92	Bin 1	17	6.333333333	98.33333333
5	95	Bin 1	20	3.333333333	98.33333333
6	99	Bin 2	0.666666667	22.66666667	98.33333333
7	104	Bin 2	5.666666667	17.66666667	98.33333333
8	110	Bin 2	11.66666667	11.66666667	98.33333333
9	118	Bin 2	19.66666667	3.666666667	121.6666667
10	122	Bin 3	0.3333333333	23	121.6666667
11	130	Bin 3	8.333333333	15	121.6666667
12	145	Bin 3	23.33333333	0	145

Q5: Standardization of Insurance Claim Amounts

S.No	laim Amount (₹ '000	x - mean	(x-mean)^2	-score = (x-mean)/std
1	12	-12.5	156.25	-1.485046613
2	15	-9.5	90.25	-1.128635426
3	18	-6.5	42.25	-0.772224239
4	20	-4.5	20.25	-0.5346167808
5	22	-2.5	6.25	-0.2970093227
6	25	0.5	0.25	0.05940186453
7	28	3.5	12.25	0.4158130517
8	30	5.5	30.25	0.6534205099
9	35	10.5	110.25	1.247439155
10	40	15.5	240.25	1.841457801

Sum of squares ->

708.5

Mean: 24.5

Standard 8.417244205

Claims Above One Standard Deviation from the Mean ($z > 1$, i.e. value $> \text{mean} + 1 \times$

Upper Thr 32.9172442

S.No	Claim Amount	Z-score	bove 1 SD? (z>1
1	12	-1.485046613	No
2	15	-1.128635426	No
3	18	-0.772224239	No
4	20	-0.5346167808	No
5	22	-0.2970093227	No
6	25	0.0594018645	No
7	28	0.4158130517	No
8	30	0.6534205099	No
9	35	1.247439155	Yes
10	40	1.841457801	Yes